

Verification of Temperley Recurrences

Summary of `verify_recurrence.py`

(a) Theory

Temperley's tooth generating functions

The paper studies generating functions

$$h_r(q),$$

where r denotes the width of a tooth and the coefficient of q^N counts teeth containing N cubes.

In his 1952 paper, Temperley derived a recurrence relation for these generating functions. The recurrence may be written in two equivalent forms.

Temperley's original multi-term recurrence

Temperley's equation (8) states that

$$(1 - q^r)h_r(q) = q^r \left(1 + \sum_{k=1}^{r-1} (r - k + 1)h_k(q) \right).$$

This expresses h_r in terms of all lower-width generating functions.

The simplified two-term recurrence

A considerably simpler recurrence, given in Temperley's note added in proof, is

$$(1 - q^r)h_r(q) = 2q h_{r-1}(q) - q^2 h_{r-2}(q).$$

This relation depends only on the two preceding generating functions and forms the computational backbone of the paper.

Explicit low-order solutions

Temperley also listed explicit expressions for the first few generating functions:

$$\begin{aligned} h_1(q) &= \frac{q}{1 - q}, \\ h_2(q) &= \frac{q^2}{(1 - q)^2}, \\ h_3(q) &= \frac{q^3(1 + q)^2}{(1 - q)(1 - q^2)(1 - q^3)}. \end{aligned}$$

These provide useful reference points for checking any implementation.

(b) What the script does

The script performs a sequence of symbolic verification tests on the generating functions $h_r(q)$.

Specifically it:

1. Computes $h_r(q)$ recursively using the two-term recurrence.

2. Extracts the numerator polynomials $N_r(q)$ appearing in

$$h_r(q) = \frac{N_r(q)}{(q; q)_r}.$$

3. Verifies agreement with Temperley's published formulas for h_1 , h_2 , and h_3 .

4. Verifies the two-term recurrence directly.

5. Verifies the original multi-term recurrence directly.

6. Demonstrates symbolically how the two-term recurrence follows from the multi-term one through cancellation of the q^r -dependent terms.

(c) How the script does it

Construction of $h_r(q)$

`compute_h(max_r)` generates the functions recursively from

$$h_r(q) = \frac{2q h_{r-1}(q) - q^2 h_{r-2}(q)}{1 - q^r}.$$

The rational expressions are simplified using SymPy's exact symbolic algebra.

Closed-form factorisation

For each computed $h_r(q)$, the script multiplies by

$$(q; q)_r = \prod_{k=1}^r (1 - q^k)$$

and factors the result to obtain the numerator polynomial

$$N_r(q).$$

The output therefore displays the representation

$$h_r(q) = \frac{N_r(q)}{(q; q)_r},$$

which is the form used throughout the paper.

Verification against Temperley (1952)

The script stores Temperley's published expressions for

$$h_1, h_2, h_3,$$

and checks that

$$h_r^{\text{computed}}(q) - h_r^{\text{Temperley}}(q) = 0$$

symbolically after simplification.

Because the calculation is exact, agreement is proved rather than estimated numerically.

Verification of the two-term recurrence

For each

$$r = 2, \dots, 7,$$

the script computes

$$(1 - q^r)h_r(q)$$

and

$$2q h_{r-1}(q) - q^2 h_{r-2}(q),$$

then verifies that their difference simplifies identically to zero.

Verification of the multi-term recurrence

For each

$$r = 1, \dots, 7,$$

the script computes

$$(1 - q^r)h_r(q)$$

and

$$q^r \left(1 + \sum_{k=1}^{r-1} (r - k + 1) h_k(q) \right),$$

and again verifies symbolic equality.

Derivation of the two-term recurrence

The script defines

$$F_r = (1 - q^r)h_r(q)$$

and evaluates

$$F_r - 2qF_{r-1} + q^2F_{r-2}.$$

It then checks the identity

$$F_r - 2qF_{r-1} + q^2F_{r-2} = q^r(2h_{r-1} - h_{r-2}).$$

Finally, the individual terms are expanded explicitly to show that all contributions proportional to q^r cancel exactly, leaving

$$(1 - q^r)h_r = 2q h_{r-1} - q^2 h_{r-2}.$$

(d) Output produced

Closed-form generating functions

The script prints the first several generating functions in the form

$$h_r(q) = \frac{N_r(q)}{(q; q)_r}.$$

This allows direct comparison with the formulas appearing in the paper.

Verification against Temperley's equation (9)

For $r = 1, 2, 3$, the script reports whether the computed expressions match Temperley's published values.

A successful run produces

```
h_1: matches Temperley eq(9): True
h_2: matches Temperley eq(9): True
h_3: matches Temperley eq(9): True
```

Two-term recurrence verification

For each $r = 2, \dots, 7$, the script prints whether

$$(1 - q^r)h_r = 2q h_{r-1} - q^2 h_{r-2}$$

holds exactly.

Multi-term recurrence verification

For each $r = 1, \dots, 7$, the script prints whether Temperley's original recurrence is satisfied.

Derivation check

The script prints the result of the symbolic cancellation argument and confirms that the cross-terms proportional to q^r vanish identically.

(e) What we learn

The implementation reproduces Temperley's original results exactly. The generating functions obtained from the recurrence agree symbolically with the explicit formulas published in Temperley's 1952 paper.

The multi-term and two-term recurrences are equivalent. Both recurrences are verified independently and shown to generate the same family of generating functions.

The simplified recurrence is mathematically justified. The symbolic derivation demonstrates precisely how the two-term recurrence emerges from the more complicated multi-term relation through cancellation of the q^r -dependent contributions.

The generating-function framework is internally consistent. Every verification is performed symbolically rather than numerically, so agreement is exact and not subject to rounding or truncation error.

The script functions as an algebraic validation tool. Rather than generating new results, it confirms the correctness of the recurrence relations, their derivation, and their implementation, providing a rigorous consistency check for the theoretical development presented in the paper.